Index of First Occurrence in a String

1. Aim

To write a C program to find the index of the first occurrence of a given character in a string and analyze its time complexity.

2. Theory

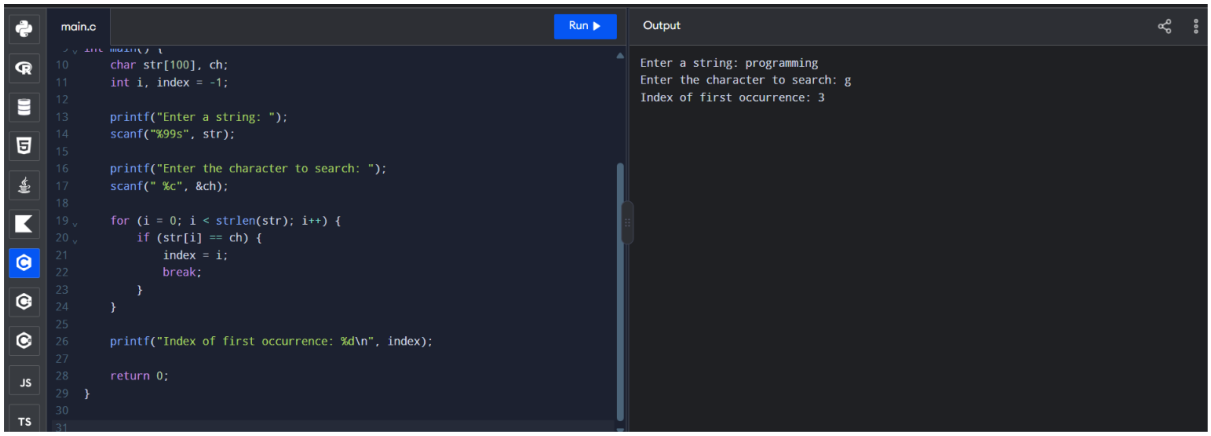
The index of the first occurrence refers to the position at which a specified character appears for the first time in a string. In C, string indexing starts from 0.

The algorithm uses a linear search technique. It checks each character from left to right and compares it with the target character. When a match is found, the algorithm returns its index and stops searching. If the character does not appear in the string, it returns -1.

For example, in the string programming, the first occurrence of the character g is at index 3.

3. Algorithm

1. Start.
2. Read a string and the target character.
3. Traverse the string from index 0.
4. Compare each character with the target character.
5. If a match is found, display its index and stop.
6. If the end of the string is reached without a match, display -1.
7. Stop.



6. Complexity Analysis

Let \(n\) be the length of the string.

- Best case: \(O(1)\), when the target character is at index 0.
- Average case: \(O(n)\), when several characters must be checked.
- Worst case: \(O(n)\), when the character is at the end or is absent.
- Space complexity: \(O(n)\) including storage for the input string; the search uses \(O(1)\) auxiliary space.

7. Result

The C program successfully finds the index of the first occurrence of a specified character in a string using linear search.